

NoisiQ: Noise-Aware Quantum Circuit Simulation and Visualization

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Abstract

In the Noisy Intermediate-Scale Quantum (NISQ) era, circuit performance is constrained by overall error rates and the specific physical characteristics of the underlying noise. NoisiQ is an open-source Python library designed to bridge the gap between quantum circuit simulators and hardware behavior through noise-aware simulation and visualization. The package dynamically routes between scalable Clifford/Pauli propagation using the stabilizer formalism and exact density-matrix simulation to model a broad suite of errors, including Pauli noise, T1 relaxation, T2 dephasing, coherent over-rotations, and correlated multi-qubit errors. NoisiQ translates these noise parameters into visual diagnostics, such as interactive heatmaps, step-by-step error propagation animations, and multi-shot error summaries, allowing users to track how errors accumulate across gates, qubits, measurements, and circuit depth. The library also provides built-in noise suppression techniques, including dynamical decoupling, while integrating with established frameworks such as Qiskit and Stim. Designed initially as an educational and exploratory research tool, NoisiQ helps early researchers connect physical noise models, circuit-level behavior, and representative performance metrics in a single workflow to easily visualize and build intuition for quantum algorithms.

1 Introduction

2 Design Philosophy

2.1 Modularity and Expansion

2.2 Teaching Tool

3 Software Architecture

3.1 Quantum Simulator

STIM

Qiskit

3.2 Noise Models

Hardware Noise

3.3 QASM

3.4 Visualization

Heatmap

Interactive Panel

Animation

4 NoisiQ Examples

4.1 Meier-Eastin-Knill (MEK) State Distillation

4.2 Greenberger-Horne-Zeilinger (GHZ) State

4.3 Dynamical Decoupling Sequences

Hardware Noise Comparison

4.4 Quantum Fourier Transform

5 Metrics

6 Conclusion

Future Work

Acknowledgments

Supporting Examples / Code

Grover's Algorithm

Deutsch-Jozsa Algorithm

Quantum Phase Estimation